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Conveyance apparatus, substrate processing apparatus, conveyance method, and article manufacturing method

Abstract

A conveyance apparatus connected to a substrate processing apparatus configured to bring a pressing member and a material on a substrate into contact with each other to form a cured film of the material on the substrate, the conveyance apparatus includes an acquisition unit configured to acquire a state of a peripheral portion of an adhesion substrate including the pressing member and the substrate adhering to the pressing member with the material interposed in between, a position adjusting unit configured to adjust a position of the adhesion substrate based on a result acquired by the acquisition unit, and a conveyance unit configured to convey the adhesion substrate adjusted by the position adjusting unit.

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Background/Summary

BACKGROUND

Field of the Disclosure

(1) The present disclosure relates to a conveyance apparatus, a substrate processing apparatus, a conveyance method, and an article manufacturing method.

Description of the Related Art

(2) As a technique capable of producing a micro structured device according to a design rule on the nanometer order and suitable for mass production, an imprint technique has been put into practical use. The imprint technique is a technique in which a mold (also referred to as a template) having a depression-protrusion structured pattern on a nanometer-scale that is formed using an electron beam drawing apparatus, an exposure apparatus, or the like is brought into contact with a moldable material (an imprint material) on a substrate to transfer the pattern. A photo-curing method is one example of the imprint technique. An imprint apparatus employing the photo-curing method molds, using a mold, a photo-curable imprint material supplied in a shot region on a substrate, cures the imprint material by light irradiation, and separates the mold from the cured imprint material, thereby forming a pattern on the substrate.

(3) In addition, conventionally, there is known a technique (a planarization technique) for planarizing irregularities of a substrate by forming a coating film on the substrate using a coating

applicator such as a spin coater. However, the planarization technique (a substrate processing apparatus) using the coating applicator may be insufficient to planarize irregularities of a substrate on a nanoscale. Thus, in recent years, it has been discussed to planarize a substrate using the imprint technique. Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2011-529626 discusses making an improvement in accuracy of planarization by dropping a composition based on irregularities of a substrate, and curing the composition in a state where a plate is in contact with the dropped composition.

(4) The planarization using the imprint technique can include a contact process of bringing a plate serving as a pressing member into contact with a composition on a substrate, a curing process of curing the composition in a contact state by light irradiation, and a release process of separating the plate from the cured composition. In the release process, it is necessary to apply a force (a release force) to separate the plate from the composition on the substrate, but the release may fail because of a large adhesive force. In a case where the release fails, it is conceivable that an adhesion substrate in which the plate and the substrate adhere to each other is carried out from the apparatus and moved to a holding housing, and a restoration process of performing the release using a release assisting mechanism using a push pin is performed.

(5) Further, in the contact process, the substrate and the plate are brought into contact with each other to align the center of the substrate and the center of the plate, but the substrate and the plate may come into contact while the centers are misaligned. If the release fails in such a state where the centers are misaligned, it is necessary to perform the restoration process on the adhesion substrate having external dimensions larger than external dimensions of the plate or the substrate. In other words, the plate and the substrate may be damaged by a collision during the movement to the holding housing, or during the release process by the release assisting mechanism.

SUMMARY

(6) According to an aspect of the present disclosure, a conveyance apparatus connected to a substrate processing apparatus configured to bring a pressing member and a material on a substrate into contact with each other to form a cured film of the material on the substrate, the conveyance apparatus includes an acquisition unit configured to acquire a state of a peripheral portion of an adhesion substrate including the pressing member and the substrate adhering to the pressing member with the material interposed in between, a position adjusting unit configured to adjust a position of the adhesion substrate based on a result acquired by the acquisition unit, and a conveyance unit configured to convey the adhesion substrate adjusted by the position adjusting unit.

(7) Further features of the present disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating a configuration of a planarization apparatus (a substrate processing apparatus).

(2) FIGS. 2A, 2B, and 2C are diagrams illustrating a planarization process.

(3) FIGS. 3A, 3B, and 3C are diagrams illustrating an imprint process.

(4) FIG. 4 is a diagram illustrating a conveyance apparatus.

(5) FIGS. 5A, 5B, 5C, 5D, 5E, and 5F are diagrams illustrating states where a substrate and a plate adhere to each other.

(6) FIGS. 6A, 6B, and 6C are diagrams illustrating a configuration of a position adjusting unit and acquired measurement values.

(7) FIG. 7 is a flowchart illustrating operation of conveying an adhesion substrate.

- (8) FIGS. **8A** and **8B** are diagrams each illustrating an example in which an adjustment is performed to enable carry-in to a holding housing.
- (9) FIGS. **9A**, **9B**, and **9C** are diagrams each illustrating a configuration of a displacement meter and acquired data.
- (10) FIG. **10** is a flowchart illustrating operation of conveying an adhesion substrate.
- (11) FIGS. **11A**, **11B**, and **11C** are diagrams illustrating a configuration of an image acquisition device and acquired images.
- (12) FIGS. **12A**, **12B**, **12C**, and **12D** are diagrams illustrating a release assisting mechanism.
- (13) FIG. **13** is a flowchart illustrating operation of conveying an adhesion substrate.

DESCRIPTION OF THE EMBODIMENTS

- (14) Exemplary embodiments will be described in detail below with reference to the attached drawings. The following embodiments are not intended to limit the present disclosure set forth in the claims. A plurality of features is described in the embodiments, but not all of the features are necessarily required for the present disclosure, and the plurality of features may be combined as desired. Furthermore, in the attached drawings, identical or similar configurations are denoted by the same reference number, and the description thereof will not be repeated.
- (15) FIG. **1** is a schematic diagram illustrating a substrate processing apparatus **100** according to a first embodiment. The substrate processing apparatus **100** (a molding apparatus) is embodied by a molding apparatus that molds a composition on a substrate **1**, using a pressing member **9**.
- (16) A specific example is a planarization apparatus that cures the composition in a state where the composition on the substrate **1** and the pressing member **9** (also referred to as the plate or super straight) having a flat surface are in contact with each other, and forms a flat layer of the composition on the substrate **1** by separating the plate from the cured composition. Another specific example is an imprint apparatus that cures the composition in a state where the pressing member **9** (also referred to as the mold) having a pattern structure is in contact with the composition on the substrate **1**, and forms a pattern structure using the cured composition. The first embodiment will be described below using the example of the planarization apparatus as the substrate processing apparatus **100**.
- (17) In the present specification and the drawings, directions are indicated in an X-Y-Z coordinate system in which a horizontal plane is an XY plane. Typically, the substrate **1** that is a processing target object is placed on a substrate stage **3** so that a surface thereof is parallel to the horizontal plane (the XY plane). Thus, hereinafter, directions orthogonal to each other in a plane along the surface of the substrate **1** are an X-axis and a Y-axis, and a direction perpendicular to the X-axis and the Y-axis is a Z-axis. Further, hereinafter, directions parallel to the X-axis, the Y-axis, and the Z-axis in the X-Y-Z coordinate system will be referred to as an X direction, a Y direction, a Z direction, respectively, and a rotation direction around the X-axis, a rotation direction around the Y-axis, and a rotation direction around the Z-axis will be referred to as a θ X direction, a θ Y direction, and a θ Z direction, respectively.
- (18) The planarization apparatus **100** cures the composition in a state where a material on the substrate **1** and the plate **9** are in contact with each other, and separates (releases) the plate **9** from the cured composition, thereby forming a planarized layer of the material on the substrate **1**.
- (19) For example, the substrate **1** can be a silicon wafer, but is not limited thereto. The substrate **1** can be made of a material freely selected from aluminum, a titanium-tungsten alloy, an aluminum-silicon alloy, an aluminum-copper-silicon alloy, silicon oxide, and silicon nitride. As the substrate **1**, a substrate on which an adhesion layer is formed by a surface treatment such as a silane coupling treatment, a silazane treatment, or formation of an organic thin film to improve the adhesion property with respect to the composition may be used. The substrate **1** typically is circular and has a diameter of 300 mm, but is not limited thereto.
- (20) The plate **9** can be made of a light transmitting material in consideration of a light irradiation process. Examples of such a material include glass, quartz, a light transmissive resin such as

polymethyl methacrylate (PMMA) or a polycarbonate resin, a transparent metal vapor deposition film, a flexible film such as dimethylpolysiloxane, a photocurable film, and a metal film. Although it is desirable that the plate **9** have a circular shape having a diameter greater than 300 mm and smaller than 500 mm, the shape of the plate **9** is not limited thereto. In addition, the thickness of the plate **9** is desirably greater than or equal to 0.25 mm and less than 2 mm, but the thickness thereof is not limited thereto.

(21) The composition can be a curable composition that is cured by being irradiated with light, e.g., an ultraviolet (UV) curable liquid. As the UV curable liquid, typically, a monomer such as an acrylate or a methacrylate can be used. The curable composition may also be referred to simply as a moldable material. Hereinafter, the moldable material will also be referred to simply as a “material”.

(22) The planarization apparatus **100** includes a substrate chuck **2**, the substrate stage **3**, a base platen **4**, columns **5**, a top plate **6**, a guide bar **7**, columns **8**, an image acquisition device **10**, a plate chuck **11**, a head **12**, and an alignment shelf **13**. The planarization apparatus **100** further includes a pressure adjustment unit **15**, a supply unit **17**, a conveyance unit **18**, an alignment scope **19**, a light source **20**, a stage drive unit **21**, an input unit **23**, and a control apparatus **200**. The substrate chuck **2** and the substrate stage **3** can move the substrate **1** while holding the substrate **1**. The plate chuck **11** and the head **12** can move the plate **9** while holding the plate **9**.

(23) The substrate **1** is carried in from the outside of the planarization apparatus **100** by the conveyance unit **18** (a conveyance apparatus) including a conveyance hand, and is held by the substrate chuck **2**. The substrate stage **3** is supported by the base platen **4**, and driven in the X direction and the Y direction to position the substrate **1** held by the substrate chuck **2** at a predetermined location. The stage drive unit **21** includes a linear motor and an air cylinder, and drives the substrate stage **3** at least in the X direction and the Y direction, but may have a function of driving the substrate stage **3** in directions of two or more axes (e.g., six axial directions). The stage drive unit **21** includes a rotation mechanism, and can rotate the substrate chuck **2** or the substrate stage **3** in the θZ direction.

(24) The plate **9** serving as the pressing member is also carried in from the outside of the planarization apparatus **100** by the conveyance unit **18** including the conveyance hand, and is held by the plate chuck **11**. The plate **9** has, for example, a circular or rectangular external shape, and has a first surface including a flat surface **25** (to be described with reference to FIG. 2A) to be in contact with the material on the substrate **1**, and a second surface opposite to the first surface. In the present embodiment, the flat surface **25** has a size equal to or slightly larger than the size of the substrate **1**. The plate chuck **11** is supported by the head **12**, and can have a function of correcting a position in the θZ direction (an inclination around the Z-axis) of the plate **9**. Each of the plate chuck **11** and the head **12** includes an opening for allowing light (ultraviolet ray) emitted from the light source **20** via a collimator lens to pass therethrough. The plate chuck **11** functions as a holding unit that mechanically holds the plate **9**. For example, the plate chuck **11** holds the plate **9** by attracting the second surface in a state where the second surface of the plate **9** faces upward. Further, the head **12** mechanically holds the plate chuck **11**. The head **12** forms a drive mechanism (not illustrated) for determining a distance between the substrate **1** and the plate **9** when bringing the plate **9** into contact with the material on the substrate **1** and separating the plate **9** from the material, and moves the plate **9** in the Z direction. For example, the drive mechanism formed by the head **12** can include an actuator such as a linear motor, an air cylinder, or a voice coil motor. Further, a load cell for measuring a pressing force (an imprint force) of the plate **9** applied to the material on the substrate **1** can be disposed in the plate chuck **11** or the head **12**. First, a plate deforming mechanism (a plate deforming unit) includes a closing member **14** for forming a closed space, using a space area A formed by a space present inside the plate chuck **11** and an internal space surrounded by the plate **9**. The plate deforming mechanism further includes the pressure adjustment unit **15** disposed outside the plate chuck **11** to adjust the pressure in the space area A.

The closing member **14** is formed of a light transmitting flat plate member such as quartz glass, and includes a connection port (not illustrated) of a pipe **16** connected to the pressure adjustment unit **15**, in a part thereof. The pressure adjustment unit **15** can increase an amount of deformation of the plate **9** protruding toward the substrate **1** by increasing the pressure in the space area A. The pressure adjustment unit **15** can also decrease the amount of deformation of the plate **9** protruding toward the substrate **1** by reducing the pressure in the space area A. The columns **5** supporting the top plate **6** are disposed on the base platen **4**. The guide bar **7** is suspended from the top plate **6**, penetrates the alignment shelf **13**, and is fixed to the head **12**. The alignment shelf **13** is suspended from the top plate **6** with the columns **8** interposed in between. The alignment shelf **13** is penetrated by the guide bar **7**. In addition, for example, a displacement meter **24** for measuring the height (flatness) of the substrate **1** held by the substrate chuck **2**, using an oblique incident image shift method, is disposed on the alignment shelf **13**.

(25) The alignment scope **19** includes an optical system and an image pickup system for observing a reference mark on the substrate stage **3** and an alignment mark on the plate **9**. However, in a case where the alignment mark is not formed on the plate **9**, the alignment scope **19** may not be provided. The alignment scope **19** is used in an alignment where the alignment scope **19** measures a relative position between the reference mark on the substrate stage **3** and the alignment mark on the plate **9** and corrects a positional shift obtained thereby.

(26) The supply unit **17** has a dispenser including a discharge outlet (a nozzle) for discharging the material in an uncured state to the substrate **1**, and supplies (applies) the material onto the substrate **1**. The supply unit **17** employs a method such as a piezo jet method or a micro solenoid method, and can supply a material having a minute volume of about 1 picoliter (pL) to the substrate **1**. The number of discharge outlets in the supply unit **17** is not limited and may be one (a single nozzle), or a plurality of (e.g., 100 or more) nozzles may be used. The plurality of nozzles may form a linear nozzle array in one row or in a plurality of rows.

(27) The image acquisition device **10** acquires an image including a region where the plate **9** and the material on the substrate **1** are in contact with each other. The image acquisition device **10** is located above the head **12**, i.e., upstream in the irradiation direction of the light source **20**. The image acquisition device **10** is, for example, an image pickup apparatus such as a charge coupled device (CCD) camera, and can acquire information representing an image of a region. The image acquisition device **10** also acquires an image of a region including the substrate **1** and the substrate chuck **2**. An image acquisition device different from the image acquisition device **10** may be provided to acquire the image. An image processing device **201** (a processing unit) processes an image and detects the position of an object in the image. The image acquired by the image acquisition device **10** is processed by the image processing device **201**, so that the state of a region where the plate **9** and the material on the substrate **1** are in contact with each other can be detected.

(28) The control apparatus **200** includes a computer apparatus including a central processing unit (CPU) and a memory, and controls the entire planarization apparatus **100**. The control apparatus **200** functions as a processing unit that performs a planarization process by comprehensively controlling each unit of the planarization apparatus **100**. Here, the planarization process is a process of planarizing the material by bringing the flat surface **25** of the plate **9** into contact with the material on the substrate **1** to make the flat surface **25** replicate the surface profile of the substrate **1**. In general, the planarization process is performed lot by lot, i.e., for each of a plurality of substrates included in the same lot.

(29) Next, the planarization process by the planarization apparatus **100** will be described with reference to FIGS. 2A, 2B, and 2C. First, a material **28** is supplied by the supply unit **17** to the substrate **1** where a base pattern **1a** is formed. FIG. 2A illustrates a state after the material **28** is placed on the substrate **1** and before the plate **9** is brought into contact with the material **28**. Next, as illustrated in FIG. 2B, the material **28** on the substrate **1** and the flat surface **25** of the plate **9** are brought into contact with each other. The plate **9** presses the material **28** so that the material **28**

spreads over the entire surface of the substrate **1**. FIG. 2B illustrates a state where the entire surface of the flat surface **25** of the plate **9** is in contact with the material **28** on the substrate **1**, and the flat surface **25** of the plate **9** replicates the surface profile of the substrate **1**. Subsequently, in the state illustrated in FIG. 2B, the material **28** on the substrate **1** is irradiated with light from the light source **20** via the plate **9**, so that the material **28** is cured. Afterward, the plate **9** is separated from the cured material **28** on the substrate **1**. A cured layer (a planarized layer) of the material **28** having a uniform thickness is thereby formed on the entire surface of the substrate **1**. FIG. 2C illustrates a state where the planarized layer of the material **28** is formed on the substrate **1**. Hereinafter, the contact (adhesion) and the separation between the flat surface **25** of the plate **9** and the material **28** on the substrate **1** will be referred to simply as the contact (adhesion) and the separation between the plate **9** and the material **28** on the substrate **1**, respectively.

(30) The planarization process of the planarization apparatus **100** has been described with reference to FIGS. 2A, 2B, and 2C. However, because the substrate processing apparatus according to the present disclosure is also applicable to the imprint apparatus that forms the pattern on the substrate, an imprint process by the imprint apparatus will also be described with reference to FIGS. 3A, 3B, and 3C.

(31) FIG. 3A illustrates a state after the material **28** is supplied onto the substrate **1** and before a mold serving as the pressing member **9** is brought into contact with the material **28**. The mold **9** has a pattern surface **26**, and a pattern structure P is formed in a pattern area of the pattern surface **26**. The pattern structure P can be composed of depressions and protrusions formed on the pattern surface **26**. Then, as illustrated in FIG. 3B, the substrate **1** to which the material **28** has been applied and the mold **9** having the pattern structure P of the pattern surface **26** are brought into contact with each other. FIG. 3B illustrates a state where the entire surface of the mold **9** having the pattern structure P is in contact with the material **28** on the substrate **1**, and the pattern surface **26** of the mold **9** has replicated the surface profile of the substrate **1**. Subsequently, in the state illustrated in FIG. 3B, the material **28** on the substrate **1** is irradiated with light from the light source **20** via the mold **9**, so that the material **28** is cured. Then, the mold **9** is separated from the cured material **28** on the substrate **1** (a release process). A cured layer having a pattern structure can be thereby formed on the entire surface of the substrate **1**. FIG. 3C illustrates a state where the pattern of the material **28** is formed on the substrate **1**. Hereinafter, the contact (adhesion) and the separation between the pattern surface **26** of the mold **9** and the material **28** on the substrate **1** will be referred to simply as the contact (adhesion) and the separation between the mold **9** and the material **28** on the substrate **1**, respectively.

(32) Next, the conveyance of the plate **9** and the substrate **1** will be described with reference to FIG. 4. FIG. 4 is a diagram illustrating an example of a configuration with regard to the conveyance unit **18**. The conveyance unit **18** may be provided to each of the plate **9** and the substrate **1**, but may be shared as illustrated in FIG. 4. In the present embodiment, the conveyance unit **18** will be described as a part of the planarization apparatus **100**, but the conveyance unit **18** may be provided as a device external to the planarization apparatus **100** to convey the plate **9** and the substrate **1** to the planarization apparatus **100**. The transfer of the plate **9** and the substrate **1** between the inside and the outside of the planarization apparatus **100** is performed via a load port **29**. The load port **29** is an interface portion for taking out and inserting the substrate **1** present inside a front opening unified pod (FOUP) used as a holding housing, from and into the planarization apparatus **100**. In the present embodiment, a plurality of plates or a plurality of substrates is stored in the FOUP.

(33) First, a method of conveying the substrate **1** by the conveyance unit **18** will be described. A conveyance hand **30** holds the substrate **1** and conveys the substrate **1**. The conveyance hand **30** can include an edge clamp that uses a pad for holding the substrate **1**, or vacuum suction. The conveyance hand **30** acquires the substrate **1** carried in from the FOUP via the load port **29** and conveys the acquired substrate **1** to a position adjusting unit **33**. The position adjusting unit **33**

adjusts the central position of the substrate **1** and adjusts the alignment of the substrate **1** in the rotation direction so that a sending-in position when conveying the substrate **1** to the substrate chuck **2** of the substrate stage **3** is always constant. The position adjusting unit **33** includes a drive stage **38** and a substrate holding mechanism **40** (both to be described with reference to FIG. **6A**), and an acquisition unit. The drive stage **38** has a drive mechanism for each of the X, Y, Z, θX , θY , and θZ directions.

(34) The substrate holding mechanism **40** holds the substrate **1** by suction using a substrate suction pad or the like. Vacuum suction, electrostatic suction, and other suction method may be used for the suction. For the positioning adjustment (pre-alignment) of the substrate **1**, the adjustment can be performed using the acquisition unit that can acquire the position of the substrate **1**. A specific configuration of the acquisition unit can be implemented by a measurement unit **37** (to be described with reference to FIG. **6A**) that can acquire position information about a peripheral portion of the substrate **1** while rotating the substrate **1**. The acquisition unit may include an image pickup device such as a CCD camera, and a processor that detects the position of an object on an image by processing the image.

(35) The substrate **1** whose position is adjusted by the position adjusting unit **33** is acquired by the conveyance hand **30**, and the acquired substrate **1** is conveyed to the substrate chuck **2** in the planarization apparatus **100**. Subsequently, the substrate **1** is subjected to the planarization process in the planarization apparatus **100**. Afterward, the substrate **1** is acquired using the conveyance hand **30** from the substrate chuck **2**, and the acquired substrate **1** is carried out to the FOUP of the load port **29**.

(36) Next, a method of conveying the plate **9** using the conveyance unit **18** will be described. The conveyance hand **30** acquires the plate **9** carried in from the FOUP via the load port **29**, and conveys the acquired plate **9** to a reversing unit **31**. The reversing unit **31** includes a holding mechanism for holding the plate **9** and a rotation mechanism for rotating the holding mechanism 180 degrees. The reversing unit **31** further includes a holding mechanism **32** that temporarily holds the reversed plate **9**. The plate **9**, which is disposed on the holding mechanism **32** so that a downward facing surface of the plate **9** is the first surface (the front side) and an upward facing surface of the plate **9** is the second surface (the back side), is acquired by the conveyance hand **30** from the holding mechanism **32**, and the acquired plate **9** is conveyed to the position adjusting unit **33**. The position adjusting unit **33** adjusts the central position of the plate **9** and adjusts the positioning of the plate **9** in the rotation direction so that a sending-in position when conveying the plate **9** to the plate chuck **11** of the head **12** is always constant. The plate **9** whose positioning is adjusted by the position adjusting unit **33** is acquired by the conveyance hand **30**, and the acquired plate **9** is conveyed to the plate chuck **11** in the planarization apparatus **100**. Subsequently, after the planarization process is fully completed in the planarization apparatus **100**, the plate **9** is acquired by the conveyance hand **30** from the plate chuck **11**. After the plate **9** is rotated 180 degrees by the reversing unit **31**, the rotated plate **9** is carried out to the FOUP of the load port **29**. The plate chuck **11** may be removable from the head **12**, and the plate **9** may be mounted on the plate chuck **11** in the conveyance unit **18** and carried in to or carried out from the planarization apparatus in this state.

(37) A controller **34** controls operation of each of the plate chuck **11**, the reversing unit **31**, the conveyance hand **30**, the load port **29**, and the position adjusting unit **33**. The controller **34** may be implemented by the control apparatus **200**.

(38) Next, how the plate **9** and the substrate **1** overlap each other when the planarization apparatus **100** brings the plate **9** and the substrate **1** into contact with each other with the material **28** in between will be described with reference to FIGS. **5A**, **5B**, **5C**, **5D**, **5E**, and **5F**. Here, illustration of the material **28** is omitted, and there is illustrated an example in which the plate **9** having a circular shape and made of a light transmitting material in consideration of the light irradiation process is used, and a silicon substrate having also a circular shape and about the same size as the size of the plate **9** is used as the substrate **1**. A notch **N** indicating the orientation of the substrate **1** is formed

on the substrate edge of the substrate **1**.

(39) FIGS. 5A to 5C illustrate a state where the central position of the plate **9** and the central position of the substrate **1** are aligned and the plate **9** and the substrate **1** adhere to each other, and FIGS. 5D to 5F illustrate a state where the central position of the plate **9** and the central position of the substrate **1** are not aligned and the plate **9** and the substrate **1** adhere to each other. FIG. 5B is a diagram illustrating the plate **9** and the substrate **1** viewed from above the plate **9** (in the Z-axis), and a single circle is formed in a case where the central position of the plate **9** and the central position of the substrate **1** are aligned as illustrated in FIG. 5B. FIG. 5C is a cross-sectional diagram at an A-A' position in FIG. 5B. In a case where the plate **9** and the substrate **1** adhere to each other and have the respective central positions being aligned, end surface position of the plate **9** and the end surface position of the substrate **1** are aligned.

(40) FIG. 5E is a diagram illustrating the plate **9** and the substrate **1** viewed from above the plate **9** (in the Z-axis), and as illustrated in FIG. 5E, two overlapping circles are formed in a case where the plate **9** and the substrate **1** adhere to each other in a state where the respective central positions are not aligned. The plate **9** and the substrate **1** have overlapping edge positions C1 and C2. FIG. 5F is a cross-sectional diagram at a B-B' position in FIG. 5E. The end surface position of the plate **9** and the end surface position of the substrate **1** are not aligned, and a region **50** of only the plate **9** or the substrate **1** is generated at each of both end surfaces. In other words, this is a state where the external dimensions of the two overlapping circles are larger than the external dimensions of the plate **9** or the substrate **1** alone.

(41) In this way, in the planarization apparatus **100**, the plate **9** and the substrate **1** are overlapped with each other to align the respective central positions when being brought into contact with each other, but a case where the central positions are not aligned because of a positional shift or the like can occur.

(42) In addition, in the release process of the planarization apparatus **100**, it is necessary to apply a force (a release force) to separate the plate **9** from the material on the substrate **1**, but the release can fail because of a large adhesive force. In the case where the release thus fails, it is conceivable that an adhesion substrate in which the plate **9** and the substrate **1** adhere to each other is carried out from the apparatus once and moved to the FOUP, and the restoration process of performing the release using a release assisting mechanism is performed.

(43) In such a process, in a case where the plate **9** and the substrate **1** in the adhesion substrate adhere to each other in the state where the respective central positions are not aligned as illustrated in FIGS. 5D to 5F, it is necessary to carry out the adhesion substrate having external dimensions larger than external dimensions of the substrate **1** or the plate **9** to the FOUP or the release assisting mechanism. However, because of the unexpectedly large external dimensions, the plate **9** and the substrate **1** can be damaged by interference with the FOUP or a member inside the apparatus during the conveyance.

(44) In the present embodiment, there will be described a method of measuring the edge position of the peripheral portion of the adhesion substrate in which the plate **9** and the substrate **1** adhere to each other and fail to separate from each other by the measurement unit **37**, and then conveying the adhesion substrate after a positional adjustment is performed, so that the adhesion substrate can be handled without interference.

(45) FIG. 7 is a flowchart illustrating operation of the conveyance unit **18** (the planarization apparatus **100**) when the release process in the planarization process is determined as having failed and an adhesion substrate **110** is carried out. The controller **34** or the control apparatus **200** comprehensively controls each unit, thereby executing the following process.

(46) In step S701, the controller **34** conveys the adhesion substrate **110**, in which the plate **9** and the substrate **1** adhere to each other and fail to separate from each other because of a strong adhesive force between the plate **9** and the substrate **1** in a release processing process, to the position adjusting unit **33** by controlling the conveyance hand **30**. The adhesion substrate **110** conveyed to

the position adjusting unit **33** is held by the substrate holding mechanism **40** of the position adjusting unit **33** as illustrated in FIG. **6A**.

(47) In step **S702**, the controller **34** rotates the adhesion substrate **110** by controlling the drive stage **38** of the position adjusting unit **33**, and acquires position information such as an overlap position and an overlap amount by measuring the state of the peripheral portion of the adhesion substrate **110** formed of the plate **9** and the substrate **1**, using the measurement unit **37** of the position adjusting unit **33**. A detailed measurement method in the measurement unit **37** will be described below.

(48) In step **S703**, based on the acquired position information, the controller **34** determines whether it is possible to carry out the adhesion substrate **110** without causing a collision to the FOUP mounted in the load port **29** that carries out without performing a positional adjustment. In a case where the adhesion substrate **110** has no positional shift as illustrated in FIG. **5A** or has a small positional shift, it is possible to determine that the adhesion substrate **110** can be carried out as it is. Specifically, the determination can be performed based on whether a positional shift amount exceeds a predetermined threshold.

(49) In a case where the controller **34** determines that it is possible to carry out the adhesion substrate **110** without a positional adjustment (YES in step **S703**), the operation proceeds to step **S706**. In step **S706**, the adhesion substrate **110** is carried out to the holding housing such as the FOUP, and the process ends. In a case where the controller **34** determines that the carry-out is not possible because interference with the FOUP or the like can occur if the adhesion substrate **110** is carried out as it is (NO in step **S703**), the operation proceeds to step **S704**. In step **S704**, the controller **34** determines whether the positional shift is a shift in a range adjustable by the position adjusting unit **33** based on the acquired position information. Specifically, the determination can be performed based on whether the positional shift amount exceeds a predetermined threshold for enabling an adjustment by the position adjusting unit **33**.

(50) In a case where the controller **34** determines that the adjustment is possible (YES in step **S704**), the operation proceeds to step **S705**. In step **S705**, the controller **34** performs a positional adjustment on the adhesion substrate **110** to establish a state where no interference occurs during carry-out to the FOUP, by moving the drive stage **38** of the position adjusting unit **33** in the X and Y directions and by rotating the drive stage **38** of the position adjusting unit **33** in a θ direction. Afterward, the operation proceeds to step **S706**. In step **S706**, the adhesion substrate **110** is carried out to the holding housing such as the FOUP, and the process ends.

(51) On the other hand, in a case where the controller **34** determines that the adjustment is not possible (NO in step **S704**), the adhesion substrate **110** cannot be safely carried out, and thus the operation proceeds to step **S707**. In step **S707**, the controller **34** stops the conveyance of the adhesion substrate **110** from the conveyance unit **18** by the conveyance hand **30**, and prompts manual carry-out by a person such as an operator.

(52) By such a process, the adhesion substrate **110** that has failed to be separated can be safely carried out from the planarization apparatus **100** (a planarization processing unit).

(53) Next, the measurement method used by the measurement unit **37** when a substrate overlapping state is determined will be described with reference to FIGS. **6A**, **6B**, and **6C**. FIG. **6A** is a diagram illustrating a configuration of the position adjusting unit **33**. The position adjusting unit **33** includes the drive stage **38**, a support portion **39**, the substrate holding mechanism **40**, and the measurement unit **37**. The drive stage **38** has the drive mechanism (not illustrated) for the X and Y directions and the θ rotation direction. The support portion **39** supports the drive stage **38**. The substrate holding mechanism **40** holds the substrate **1**. The measurement unit **37** is disposed at a position where the measurement unit **37** can measure the position of the peripheral portion of the adhesion substrate **110**, i.e., the positions of the end surface of the substrate **1** and the end surface of the plate **9**, when the adhesion substrate **110** is rotated by the drive stage **38**. The substrate **1**, which is placed on the substrate holding mechanism **40** on the drive stage **38** by the conveyance hand **30**, is held by the

substrate suction pad in the substrate holding mechanism **40**. Vacuum suction, electrostatic suction, and other suction method may be used for the suction.

(54) The drive stage **38** is rotated, and the measurement unit **37** measures the end surface position of the substrate **1** and the end surface position of the plate **9**, thereby acquiring position information about the peripheral portion of the adhesion substrate **110** (i.e., an overlap position and an overlap amount between the plate **9** and the substrate **1**) and position information about the substrate **1**. Based on the information such as the overlap position and the overlap amount acquired by the measurement unit **37**, the drive stage **38** is moved in the X and Y directions and rotated in the θ direction to avoid interference at the FOUP and during the conveyance, so that the position of the adhesion substrate **110** is adjusted. This makes it possible to prevent the adhesion substrate **110** from being damaged in a collision during the process of conveyance from the planarization apparatus **100** to the FOUP mounted in the load port **29**.

(55) FIG. **6B** is a waveform chart illustrating a signal acquired by the measurement unit **37**. When the drive stage **38** holding the adhesion substrate **110** rotates, the position of the edge (peripheral portion) of the adhesion substrate **110** in a radial direction continually changes, and thus waveform data indicated by a solid line can be obtained as the position information. With regard to the waveform data, a signal processing device (not illustrated) of the position adjusting unit **33** performs the first-order differentiation on the above-described signal, so that a signal illustrated in FIG. **6C** is obtained. Here, if the first-order differentiation signal is horizontally linear, it can be said that the central position of the substrate **1** and the central position of the plate **9** are in an aligned state. However, in a case where the plate **9** and the substrate **1** overlap each other in a shifted state as in FIG. **5D**, the position of the notch N of the substrate **1** and the overlapping edge positions C1 and C2 of the substrate **1** and the plate **9** are detected by the first-order differentiation. If a notch position of the substrate **1** is identified, the overlap position between the plate **9** and the substrate **1** can be identified based on the notch position. An overlap amount can also be calculated from a rotation amount of a distance between the positions C1 and C2. From the data obtained by the first-order differentiation, a distinction between the notch N and the overlapping edge positions C1 and C2 can be made based on a peak amount and a change section amount of each of the notch N and the overlapping edge positions C1 and C2. Using the results, the position is adjusted by the position adjusting unit **33** not to cause a collision of the substrate **1** during the carry-out, and subsequently, the carry-out is performed.

(56) FIGS. **8A** and **8B** each illustrate an example in which the adhesion substrate **110** is carried in to a FOUP **41** after the position is adjusted by the position adjusting unit **33**. In FIG. **8A**, the adhesion substrate **110** is carried in after the position thereof is adjusted by the position adjusting unit **33** rotating the adhesion substrate **110** so that the longitudinal direction of the adhesion substrate **110** is parallel with the direction of carry-in to the FOUP **41** and the central position of the adhesion substrate **110** is aligned with the central position of the FOUP **41**. In FIG. **8B**, the position thereof is adjusted by the position adjusting unit **33** rotating the adhesion substrate **110** so that the longitudinal direction of the adhesion substrate **110** intersects the direction of carry-in to the FOUP **41** and the central position of the adhesion substrate **110** is aligned with the central position of the FOUP **41**. Such an adjustment can be appropriately performed to avoid interfere with the FOUP or the like, depending on the size of the FOUP **41** and the shift amount of the adhesion substrate **110**.

(57) The first embodiment is described based on the example in which the measurement unit **37** that measures the edge position of the adhesion substrate **110** is provided in the position adjusting unit **33** as the acquisition unit. In a second embodiment, there will be described an example in which a displacement meter **24** in a measurement device **60** is used as the acquisition unit in place of the position adjusting unit **33**. In the present embodiment, a part different from the first embodiment will be mainly described, and description of a similar part will be omitted.

(58) FIG. **10** is a flowchart illustrating operation of a conveyance unit **18** (a planarization apparatus **100**) when release in a planarization process is determined to have failed and an adhesion substrate

110 is carried out. A controller **34** or a control apparatus **200** comprehensively controls each unit, thereby executing the following process.

(59) In step **S1001**, the controller **34** conveys the adhesion substrate **110** in which a plate **9** and a substrate **1** adhere to each other and fail to separate from each other because of a strong adhesive force between the plate **9** and the substrate **1** in a release processing process, to the measurement device **60** by controlling a conveyance hand **30**. The adhesion substrate **110** conveyed to the measurement device **60** is held by a substrate chuck **52** of the measurement device **60** as illustrated in FIG. **9A**.

(60) In step **S1002**, the controller **34** rotates the adhesion substrate **110** by controlling a drive stage **53** of the measurement device **60**, and acquires position information such as an overlap position and an overlap amount by measuring the state of the peripheral portion of each of the plate **9** and the substrate **1**, using the displacement meter **24** of the measurement device **60**. A detailed measurement method in the displacement meter **24** will be described below.

(61) In step **S1003**, based on the acquired position information, the controller **34** determines whether it is possible to carry out the adhesion substrate **110** without causing a collision to an FOUP mounted in a load port **29** that carries out without performing a positional adjustment. In a case where the adhesion substrate **110** has no positional shift as illustrated in FIG. **5A** or has a small positional shift, it is possible to determine that the adhesion substrate **110** can be carried out as it is. Specifically, the determination can be performed based on whether a positional shift amount exceeds a predetermined threshold.

(62) In a case where the controller **34** determines that it is possible to carry out the adhesion substrate **110** without a positional adjustment (YES in step **S1003**), the operation proceeds to step **S1007**. In step **S1007**, the adhesion substrate **110** is carried out to a holding housing such as the FOUP, and the process ends. In a case where the controller **34** determines that the carry-out is not possible because interference with the FOUP or the like can occur if the adhesion substrate **110** is carried out as it is (NO in step **S1003**), the operation proceeds to step **S1004**. In step **S1004**, the controller **34** determines whether the positional shift is a shift in a range adjustable by a position adjusting unit **33** based on the acquired position information. Specifically, the determination can be performed based on whether the positional shift amount exceeds a predetermined threshold for enabling an adjustment by the position adjusting unit **33**.

(63) In a case where the controller **34** determines that the adjustment is possible (YES in step **S1004**), the operation proceeds to step **S1005**. In step **S1005**, the controller **34** conveys the adhesion substrate **110** from the measurement device **60** to the position adjusting unit **33**.

(64) In step **S1006**, the controller **34** performs a positional adjustment on the adhesion substrate **110** to establish a state where no interference occurs during carry-out to the FOUP, by moving a drive stage **38** of the position adjusting unit **33** in X and Y directions and by rotating the drive stage **38** of the position adjusting unit **33** in a θ direction. Afterward, the operation proceeds to step **S1007**. In step **S1007**, the adhesion substrate **110** is carried out to the holding housing such as the FOUP, and the process ends.

(65) On the other hand, in a case where the controller **34** determines that the adjustment is not possible (NO in step **S1004**), the adhesion substrate **110** cannot be safely carried out, and thus the operation proceeds to step **S1008**. In step **S1008**, the controller **34** stops the conveyance of the adhesion substrate **110** from the conveyance unit **18** by the conveyance hand **30**, and prompts manual carry-out by a person such as an operator.

(66) By such a process, the adhesion substrate **110** that has failed to be separated can be safely carried out from the planarization apparatus **100** (a planarization processing unit).

(67) The measurement method using the displacement meter **24** of the measurement device **60** according to the present embodiment will be described with reference to FIGS. **9A**, **9B**, and **9C**.

(68) The measurement device **60** may be provided inside the conveyance unit **18** or may be provided outside the conveyance unit **18**. The measurement device **60** includes the substrate chuck

52, the drive stage **53**, and the displacement meter **24**. The substrate chuck **52** holds the substrate **1**. The drive stage **53** includes a drive mechanism (not illustrated) for the X and Y directions and the θ rotation direction.

(69) For example, a spectral interference laser displacement meter for spectral interferometry can be used for the displacement meter **24**.

(70) The displacement meter **24** is disposed at a position where light is emitted toward the plate **9** from an upper surface of the substrate chuck **52**. FIG. **9A** is a cross-sectional diagram illustrating the adhesion substrate **110** in which the plate **9** and the substrate **1** overlap each other and have the respective central positions being aligned as in FIG. **5A**. The plate **9** is made of a light transmitting resin. The light emitted from the displacement meter **24** reflects off each of a surface A of the plate **9** and an upper surface B of the substrate **1**, and returns to the displacement meter **24**. An optical path **42** indicates the light reflected by each of these surfaces. In a graph that indicates the intensity of light in an optical path length measured by the displacement meter **24**, the surface position of each of the plate **9** and the substrate **1** is indicated by an optical path length and a light intensity peak.

(71) FIG. **9B** is a cross-sectional diagram illustrating the adhesion substrate **110** having a positional shift because the central position of the plate **9** and the central position of the substrate **1** are not aligned as in FIG. **5B**. The light emitted from the displacement meter **24** reflects off each of the surface A of the plate **9**, an adhesion surface C of the plate **9**, and a substrate chuck surface D, and returns to the displacement meter **24**. In a graph that indicates the intensity of light in an optical path length measured by the displacement meter **24**, the position of each of the surface of the plate **9**, the adhesion surface of the plate **9**, and the surface of the substrate chuck **52** is indicated as an optical path length and a light intensity peak. Occurrence of a shift in a positional relationship between the plate **9** and the substrate **1** can be determined from the number of such peaks. FIG. **9C** illustrates measurement by the displacement meter **24** after the drive stage **53** is X-driven from the position in FIG. **9B**. In FIG. **9C**, the light emitted from the displacement meter **24** reflects off each of the surface A of the plate **9** and the upper surface B of the substrate **1**, and returns to the displacement meter **24**. An overlap amount of the adhesion substrate **110** can be identified from a change in the X-drive amount of the drive stage **53** and the number of peaks of light intensity. In a case where the spectral interference laser displacement meter is used as the displacement meter **24**, a light intensity peak position (distance) changes because of the influence of the refractive index of air and glass, and thus a reflection surface and a peak position can be shifted. However, FIGS. **9A**, **9B**, and **9C** illustrate waveforms corrected based on only the difference in refractive index depending on the material, for convenience. Further, while the upper surface B of the substrate **1** is expressed as the reflection surface for the light from the displacement meter **24**, the light can reflect off the adhesion surface C of the plate **9** because the upper surface B of the substrate **1** and the adhesion surface C of the plate **9** are substantially at the same position. However, in this measurement method, which one of these surfaces is the reflection surface does not matter. The method makes it possible to identify the position information (an overlap position and an overlap amount) about the plate **9** and the substrate **1** by measuring at least three points of the substrate peripheral portion using the displacement meter **24**. In other words, the measurement after the drive stage **53** is rotated in the θ direction is repeated, so that measurement at a plurality of positions of the adhesion substrate **110** by the displacement meter **24** can be performed.

(72) In a third embodiment, there will be described a method of acquiring position information (an overlap position and an overlap amount) between a plate and a substrate using an image acquisition device that can acquire an image of a peripheral portion of the substrate as the acquisition unit.

(73) In the present embodiment, there will be described an example of using an image acquisition device **10** used to acquire an image including a region where a plate **9** of a planarization apparatus **100** and a material on a substrate **1** are in contact with each other, but an image acquisition device separately provided in a conveyance unit **18** can also be used. FIG. **11A** is a diagram illustrating a

side view of the image acquisition device **10**. In a case where the position information about an adhesion substrate in which the plate and the substrate adhere to each other is acquired by the image acquisition device **10**, the position information is acquired in a state where the substrate **1** is adsorbed by the substrate chuck **2**. In the example in FIG. **11A**, the image acquisition device **10** acquires an image at each of a plurality of positions of a substrate edge **43** of the substrate **1** and a plate edge **44** of the plate **9**. A part imaged by the image acquisition device **10** is a measurement area **46**. The measurement area **46** is a range in which the substrate edge **43** and the plate edge **44** can be simultaneously measured.

(74) In the example in FIG. **11B**, four measurement areas **46** are illustrated, and two pairs of measurement areas **46** are located at the respective positions symmetrical about the central point of an inner peripheral portion of the substrate chuck **2**. The number and the positions of the measurement areas **46** whose images are acquired are not limited to the illustrated number and positions. The number and the positions of the measurement areas **46** may be input from a console screen (not illustrated) of a substrate processing apparatus by a user, the input number and positions may be saved in a storage unit (not illustrated) of a control apparatus **200**, and the input number and positions of the measurement areas **46** whose images are to be acquired by the image acquisition device **10** may be changed as appropriate.

(75) FIG. **11C** illustrates an enlarged view of an image of the measurement area **46** in FIG. **11B**. From the image acquired by the image acquisition device **10**, the image processing device **201** detects a circular arc of each of the substrate edge **43** and the plate edge **44** using a technique of detecting a circular arc such as the Hough transform. Subsequently, the image processing device **201** can determine a positional shift amount of the central point of each of the substrate edge **43** and the plate edge **44** with respect to the circular arc. An average value or a mean value of a plurality of correction amounts determined from images of a plurality of measurement areas, or a value determined by other statistical technique may be used as a correction amount.

(76) It is also possible to determine a positional shift amount based on a distance between a tangent to the circular arc of the substrate edge **43** and a tangent to the circular arc of the plate edge **44** after detecting the circular arc of each of the substrate edge **43** and the plate edge **44** from the images of the four measurement areas **46**. Specifically, at first, in an example of a measurement area **46-a** in FIG. **9C**, a tangent **47-a** passing through a point on the circular arc of the substrate edge **43** and in contact with the circular arc of the substrate edge **43** is determined. Next, a tangent **48-a** in contact with the circular arc of the plate edge **44** and in parallel with the tangent **47-a** is determined. Subsequently, a distance **49-a** between the tangent **47-a** and the tangent **48-a** is determined.

(77) A measurement area **46-d** is located at a position symmetrical to the position of the measurement area **46-a** with respect to the central point of the substrate chuck **2**. In the measurement area **46-d**, as with the measurement area **46-a**, a circular arc of each of the substrate edge **43** and the plate edge **44** is detected from the image acquired by the image acquisition device **10**.

(78) A tangent **47-d** of the substrate edge **43** and a tangent **48-d** of the plate edge **44** which are parallel to the tangent **47-a** or the tangent **48-a** are determined, and a distance **49-d** between the tangent **47-d** and the tangent **48-d** is determined. Subsequently, a positional shift amount $S-1$ between the substrate edge **43** and the plate edge **44** in a direction orthogonal to the tangent **47-a** can be determined from a difference between the distance **49-a** and the distance **49-d**.

(79) Next, using a similar method, a tangent **47-b** and a tangent **48-b** are determined from an image of a measurement area **46-b**, and a distance **49-b** between the tangents is determined. Similarly, a tangent **47-c** and a tangent **48-c** are determined from an image of a measurement area **46-c**, and a distance **49-c** between the tangents is determined. Subsequently, a positional shift amount $S-2$ between the substrate edge **43** and the plate edge **44** in a direction orthogonal to the tangent **47-b** can be determined from a difference between the distance **49-b** and the distance **49-c**.

(80) Further, a positional shift amount S can be determined from a combination of the positional

shift amount S-1 in the direction orthogonal to the tangent **47-a** and the positional shift amount S-2 in the direction orthogonal to the tangent **47-b**. In other words, the positional shift amount S can be determined from the distance between the tangents parallel to each other in each pair at four points on the circular arc of the substrate **1** and four points on the circular arc of the circular arc of the plate **9**.

(81) The process in and after step **S1003** in FIG. **10** is performed using the positional shift amount S thus determined, so that the adhesion substrate that fails to be separated can be safely carried out from the planarization apparatus **100** (the planarization processing unit) in the present embodiment as well.

(82) In the first embodiment to the third embodiment, the adjustment method when the adhesion substrate is conveyed to the holding housing such as the FOUP is described. In a fourth embodiment, a method of releasing an adhesion substrate using a release assisting mechanism will be described.

(83) The present embodiment will be described using an example in which a release assisting mechanism **45** is in a substrate stage **3** (see FIGS. **12A** to **12D**). However, a release device having the release assisting mechanism **45** may be provided as an external device of a planarization apparatus **100** to convey a plate **9** and a substrate **1** to the planarization apparatus **100**.

(84) FIG. **13** is a flowchart illustrating operation when release in a planarization process is determined as having failed and an adhesion substrate **110** is conveyed after a positional adjustment is performed so that the adhesion substrate **110** is released using the release assisting mechanism **45**. A control apparatus **200** comprehensively controls each unit, thereby executing the following process. The state where the release is determined as having failed may be either a case where the release fails even though release operation is performed using the release assisting mechanism **45**, or a case where the release fails after release operation is performed without using the release assisting mechanism **45**.

(85) In step **S1301**, the control apparatus **200** acquires, using an acquisition unit, the state of the peripheral portion of the adhesion substrate **110** in which the plate **9** and the substrate **1** adhere to each other and fail to separate from each other because of a strong adhesive force between the plate **9** and the substrate **1** in a release processing process, thereby acquiring position information such as an overlap position and an overlap amount.

(86) The method described in any of the first embodiment to the third embodiment may be used for the acquisition by the acquisition unit.

(87) In step **S1302**, based on the acquired position information, the control apparatus **200** determines whether the state of the adhesion substrate **110** corresponds to a state with a positional shift amount for which a release process using the release assisting mechanism **45** is possible.

(88) In step **S1303** (YES in step **S1302**), the control apparatus **200** conveys the adhesion substrate **110** to a position adjusting unit **33**. In step **S1304**, the control apparatus **200** performs a positional adjustment on the adhesion substrate **110** to enable the release process using the release assisting mechanism **45** by moving a drive stage **38** in X and Y directions and rotating the drive stage **38** in a θ direction.

(89) In step **S1305**, the control apparatus **200** conveys the adhesion substrate **110** to a position at which the release process using the release assisting mechanism **45** of a substrate chuck **2** of the substrate stage **3** is possible, by controlling a conveyance hand **30**.

(90) In step **S1306**, the control apparatus **200** performs release drive for driving a plate chuck **11** downward so that the plate chuck **11** holds the plate **9** by suction, and subsequently driving the plate chuck **11** upward while using the release assisting mechanism **45**. The plate **9** and the substrate **1** are thereby separated. Subsequently, the operation returns to a normal planarization process, and the process ends.

(91) On the other hand, in a case where the control apparatus **200** determines that the release process is not possible (NO in step **S1302**), the release process for the adhesion substrate **110** using

the release assisting mechanism **45** cannot be safely continued. Thus, in step **S1307**, the conveyance of the adhesion substrate **110** to the position adjusting unit **33** is stopped, and a process of carry-out to a FOUP is performed in a manner similar to the manner described in the first embodiment to the third embodiment. In a case where even the process of carry-out to the FOUP cannot be performed, manual conveyance by a person such as an operator is prompted.

(92) According to this method, it is possible to perform the release process using the release assisting mechanism **45** without carrying out the adhesion substrate having once failed in release to the outside of the planarization apparatus **100**, so that a risk such as attachment of a foreign substance can be reduced.

(93) Next, a release method by the release assisting mechanism **45** of the present embodiment will be described with reference to FIGS. **12A**, **12B**, **12C**, and **12D**.

(94) The release assisting mechanism **45** is included in the substrate stage **3** as a mechanism for assisting the release by reducing an adhesive force in the release process. The release assisting mechanism **45** has a function of pushing up the plate **9** from below in the release process, and is used as an assisting mechanism for reducing the adhesive force by allowing air to enter a portion between the plate and the substrate. The release assisting mechanism **45** may be a push pin in a size to fit into a notch shape of the substrate **1** and to be driven up and down, or may be a vent for blasting air upward from below.

(95) FIG. **12A** illustrates a state where the plate **9** is shifted to a great extent with respect to the substrate **1** when the substrate **1** is held on the substrate chuck **2** provided on the substrate stage **3** and the plate **9** is brought into contact with a material on the substrate **1**. FIG. **12B** is a cross-sectional diagram illustrating the substrate **1**, the plate **9**, the substrate chuck **2**, the substrate stage **3**, and the release assisting mechanism **45** at a position A-A' in FIG. **12A**. In a case where the substrate **1** and the plate **9** adhere to each other in a state where the respective centers are aligned, the release assisting mechanism **45** can apply a force for reducing the adhesive force to the plate **9** through the space of the notch. However, in a case where the substrate **1** and the plate **9** adhere to each other in a state where the respective centers are not aligned as in FIG. **12B**, a region **50** where the substrate **1** is not present is generated below the plate **9**. If the region **50** is generated in a case where the diameter of the substrate **1** and the diameter of the plate **9** are substantially the same, the plate **9** is shifted from the substrate **1**, and thus a region where the substrate **1** is not covered by the plate **9** is generated. In a case where the plate **9** is not present above the release assisting mechanism **45** as in FIG. **12B**, the plate **9** cannot be pushed up, so that the adhesive force is not reduced, and the plate **9** and the substrate **1** come off a plate holding mechanism or a substrate holding mechanism in a state where the plate **9** and the substrate **1** adhere to each other. Thus, in a case where such a situation occurs, it is desirable to perform the release process again after the positioning adjustment described with reference to FIG. **13** is performed.

(96) FIG. **12C** illustrates a position of the plate **9** and the substrate **1** conveyed again to the substrate chuck **2** provided on the substrate stage **3** by the conveyance hand **30** after the positioning adjustment described with reference to FIG. **13** is performed. FIG. **12D** is a cross-sectional diagram illustrating the substrate **1**, the plate **9**, the substrate chuck **2**, the substrate stage **3**, and the release assisting mechanism **45** at a position B-B' in FIG. **12C**. The adhesion substrate whose position has been adjusted by the position adjusting unit **33** so that the region **50** is placed above the release assisting mechanism **45** is conveyed, so that the release assisting mechanism **45** can apply the force for reducing the adhesive force to the plate **9** through the space of the notch.

Exemplary Embodiment of Article Manufacturing Method

(97) Next, a method of manufacturing an article (such as a semiconductor integrated circuit (IC) element, a liquid crystal display element, a color filter, or a micro electro mechanical system (MEMS)) using the above-described planarization apparatus will be described. The manufacturing method uses the above-described planarization apparatus, and includes a process of planarizing a composition by bringing the composition, which is disposed on a substrate (such as a wafer or a

glass substrate), into contact with a plate, a process of curing the composition, and a process of separating the composition and the plate. A planarized film is thereby formed on the substrate. Subsequently, the substrate on which the planarized film is formed is processed, for example, to have a pattern formed thereon using a lithography apparatus, and the processed substrate is processed in another known processing process, so that an article is manufactured. The other known processing process may be etching, resist separation, dicing, bonding, or packaging. (98) According to the present manufacturing method, an article higher in quality than conventional articles can be manufactured.

(99) While the present disclosure has been described with reference to embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(100) This application claims the benefit of priority from Japanese Patent Application No. 2022-026120, filed Feb. 22, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A conveyance apparatus connected to a substrate processing apparatus configured to bring a pressing member and a material on a substrate into contact with each other to form a cured film of the material on the substrate, the conveyance apparatus comprising: an acquisition unit configured to acquire information about a positional shift of the pressing member relative to the substrate in a state in which the material on the substrate is in contact with the pressing member; a position adjusting unit configured to adjust positions of the substrate and the pressing member based on a result acquired by the acquisition unit; and a conveyance unit configured to convey the substrate and the pressing member adjusted by the position adjusting unit.
2. The conveyance apparatus according to claim 1, wherein the acquisition unit acquires the information about the positional shift while rotating the substrate and the pressing member in the state in which the material on the substrate is in contact with the pressing member.
3. The conveyance apparatus according to claim 2, wherein the acquisition unit acquires the information about the positional shift by acquiring an edge position of a peripheral portion of the substrate and the pressing member in the state in which the material on the substrate is in contact with the pressing member.
4. The conveyance apparatus according to claim 2, wherein the acquisition unit acquires the information about the positional shift by emitting light from a side where the pressing member in the state in which the material on the substrate is in contact with the pressing member is located and measuring an optical path length of reflected light.
5. The conveyance apparatus according to claim 1, wherein the acquisition unit acquires an image of each of a plurality of regions including an edge of a peripheral portion of the substrate and the pressing member in the state in which the material on the substrate is in contact with the pressing member, and wherein the position adjusting unit adjusts the positions of the substrate and the pressing member based on the image of each of the plurality of regions.
6. The conveyance apparatus according to claim 1, wherein the position adjusting unit adjusts the position by performing at least one of movement and rotation in a direction parallel to a surface of the substrate and the pressing member in a case where it is determined that an adjustment of the position is necessary based on the result acquired by the acquisition unit.
7. The conveyance apparatus according to claim 1, wherein the conveyance unit carries the substrate and the pressing member, whose position is adjusted by the position adjusting unit, to a holding housing.
8. The conveyance apparatus according to claim 7, wherein the position adjusting unit adjusts the positions of the substrate and the pressing member to avoid interference between the holding

housing, and the substrate and the pressing member during carry-out to the holding housing.

9. The conveyance apparatus according to claim 1, wherein the conveyance unit carries the substrate and the pressing member to a release assisting mechanism configured to release the pressing member from the substrate, and wherein the position adjusting unit adjusts the positions of the substrate and the pressing member to enable release assistance by the release assisting mechanism.

10. The conveyance apparatus according to claim 1, wherein the substrate and the pressing member each have a circular shape.

11. A conveyance method comprising: acquiring information about a positional shift of a pressing member relative to a substrate in a state in which a material on the substrate is in contact with the pressing member; adjusting positions of the substrate and the pressing member based on a result of the acquiring; and conveying the substrate and the pressing member.

12. A substrate processing apparatus configured to bring a pressing member and a material on a substrate into contact with each other to form a cured film of the material on the substrate, the substrate processing apparatus comprising: an acquisition unit configured to acquire information about a positional shift of the pressing member relative to the substrate in a state in which the material on the substrate is in contact with the pressing member; a position adjusting unit configured to adjust positions of the substrate and the pressing member based on a result acquired by the acquisition unit; and a conveyance unit configured to convey the substrate and the pressing member adjusted by the position adjusting unit.

13. An article manufacturing method comprising: forming a cured film using the substrate processing apparatus to which the conveyance apparatus according to claim 1 is connected; and processing the substrate on which the cured film is formed, wherein an article is manufactured from the processed substrate.
